

# RFI Report for November 2024

## Executive summary

Based on the samples selected during the month of November, it seems that the RFI contamination has been relatively low, compared with that of [previous month](#). Although weak, the noise generated by MTN and Telkom downlinks are visible in the observation on the 3rd of November (see [subsection a](#)). The general trend in all samples is that the number of channels affected by RFI in the lower part of the band is the highest in autocorrelation products and lowest in long baselines cross correlation. There also seems to be an RFI emission around 700 MHz, visible in all products (see [subsection b](#), [subsection c](#) and [subsection d](#)). With a varying strength (i.e. fraction of the time flagged) between observations and their products, the noise appears to have started to be noticeable from the 10th of November onward.

Building on the analysis of autocorrelation and cross-correlation products in the 32k mode, the second segment of the report focuses on cross-correlation RFI behavior across both 32k and 4k modes for November 2024. The hourly analysis (Figure 13) shows average RFI levels at 45%, with notable peaks at hours in the early morning, and late evening, with 21:00 UTC being particularly severe (90%). Figure 14 reveals intense RFI activity in bands such as 544–592 MHz, 620–680 MHz, and 910–950 MHz, overlapping with previously identified sources. Interestingly, a new RFI source is evident in the 620–680 MHz range, consistent across both 32k and 4k modes, and warrants further investigation.

## 1. Introduction

The purpose of this document is to report the current status of RFI on-site. Given the construction work that has been going on over the last couple of months, it is therefore imperative to monitor the construction related activities on site that could potentially generate RFI signals which, in turn, contaminate the science observations. Preferably these reports are to be released on a weekly basis (or every two weeks).

## 2. Method

The flags data, specifically *ingest\_flags*, are considered in these analyses, as they can be used as proxies of RFI frequency occupancy, i.e. RFI emission as a function of frequency channels. The strategy is to select two to three observations per week and report the status of the RFI contamination on site based on the flags of each dataset (of each observation). For each observation

- Autocorrelation flags of a few *scans* corresponding to the *gain calibrator* are selected. This is due to the fact that, at the time of writing, there are some issues with accessing a big chunk of an entire dataset.
- Autocorrelation flags, which are three dimensional arrays, i.e. *time: frequency channels: correlation products*, are averaged over both time and correlation products.

- Autocorrelation flags become a one dimensional array which is plotted in order to check the RFI occupancy in the frequency channels.

The cross correlation flags are also investigated in order to see how much the RFI emission affects the correlation products which are the main interest in an interferometric observation. The procedure for the cross correlation flags is the same as that of autocorrelation ones. To further go into more detail, the flags corresponding to short and long baselines are split into two parts, and each part is plotted separately. Based on some datasets that were analyzed previously, the U-band observations are the ones that are most affected by the current RFI emissions on site. Hence, the majority of the datasets that will be inspected is in U-band, nevertheless observations in both L and S bands will be looked at too.

In what follows, it is worth noting that the vertical gray shaded areas in all plots denote the known RFI emissions in U-band.

### 3. Results

#### a. 2024, 03 November

The plots of the auto, short and long baselines cross correlations of the dataset collected on the 3rd of November are shown in Figures 1, 2 and 3 respectively. Overall, the level of contamination is relatively low, although a few channels affected by weak RFI (with a fraction of the time flagged less than 0.1) are noticed in the autocorrelation plot. The Telkom downlink generated RFI, although weak, is seen in all the products, and weak MTN downlink RFI is only visible in auto and long baseline cross correlations.

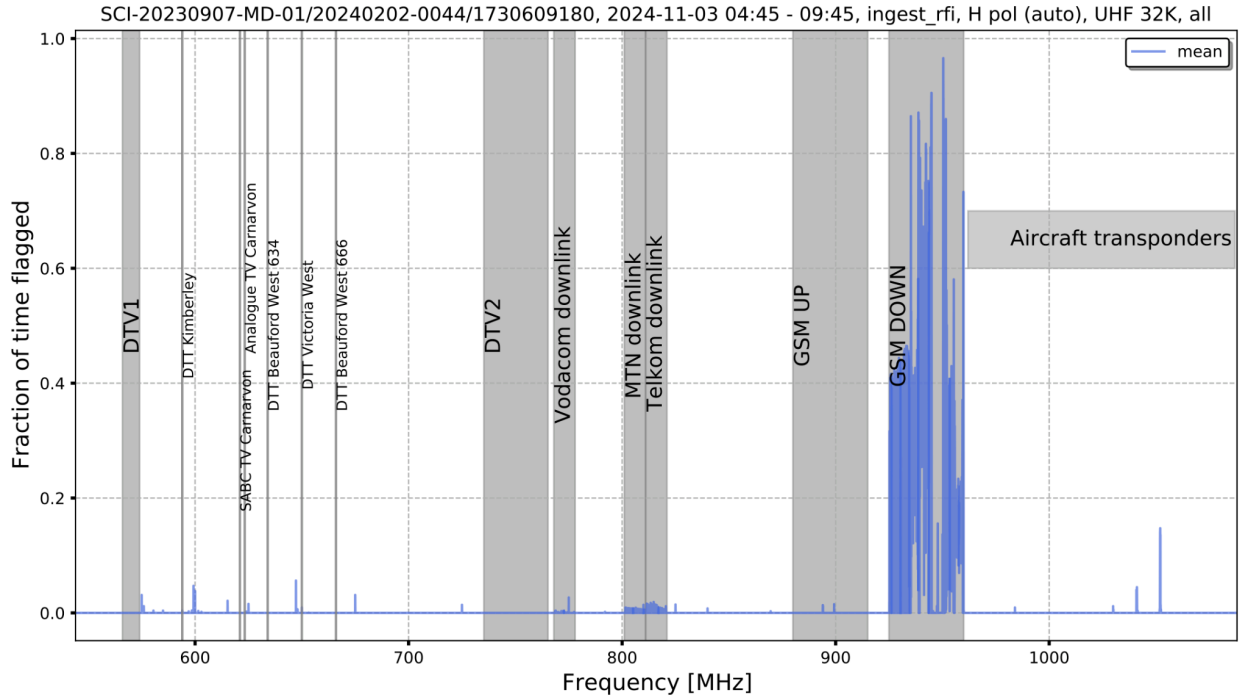


Figure 1: Mean of the autocorrelation flags of SB\_ID:20240202-0044 dataset.

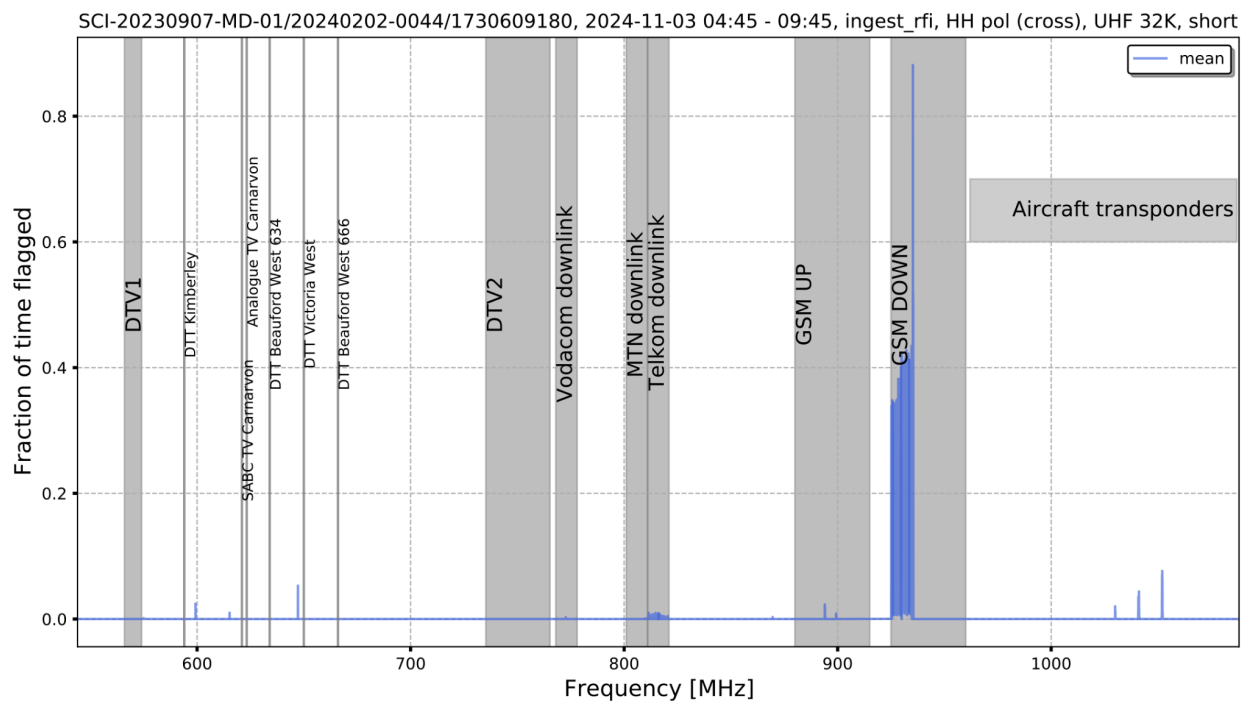


Figure 2: Mean of the cross correlation flags of the short baselines of SB\_ID:20240202-0044 dataset.

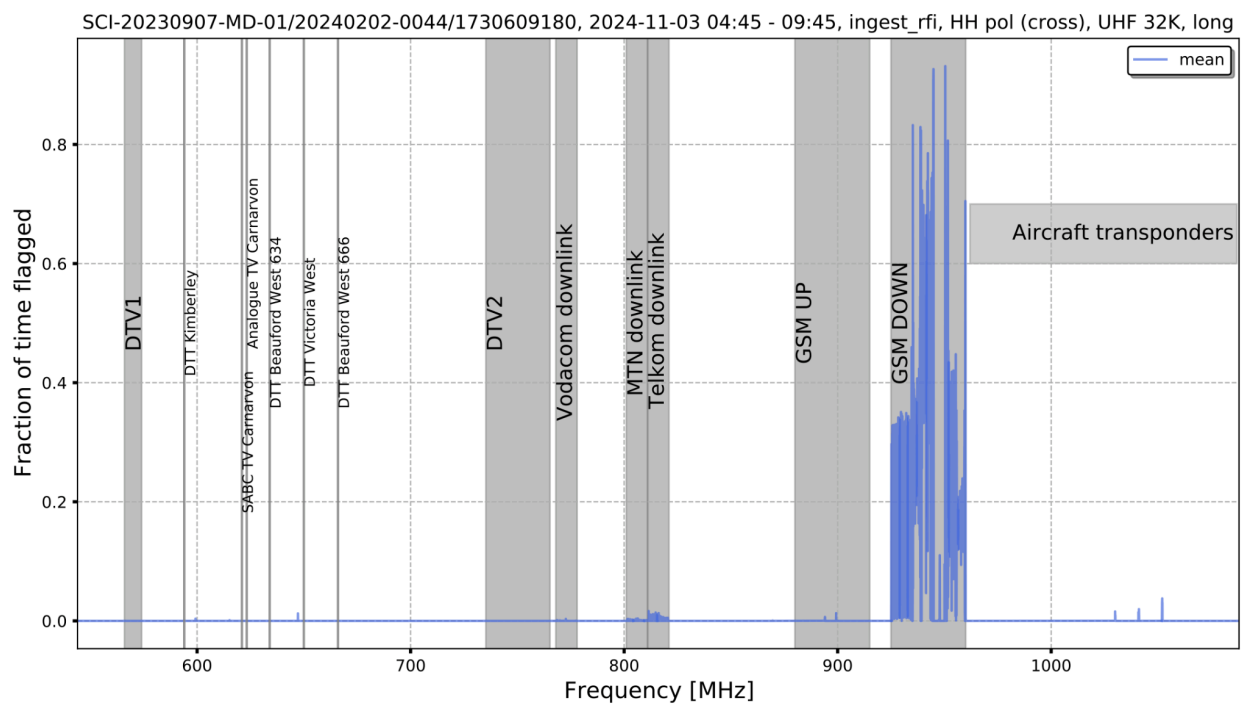


Figure 3: Mean of the cross correlation flags of long baselines of SB\_ID:20240202-0044 dataset.

**b. 2024, 10 November**

The resulting flags from the dataset collected on the 10th of November 2024, are shown in Figures 4, 5, and 6. Overall, the cross correlation is cleaner than the autocorrelation, but it seems that there is a relatively strong emission, with a fraction of the time flagged equal to  $\sim 0.2$ , close to 700 MHz in all products.

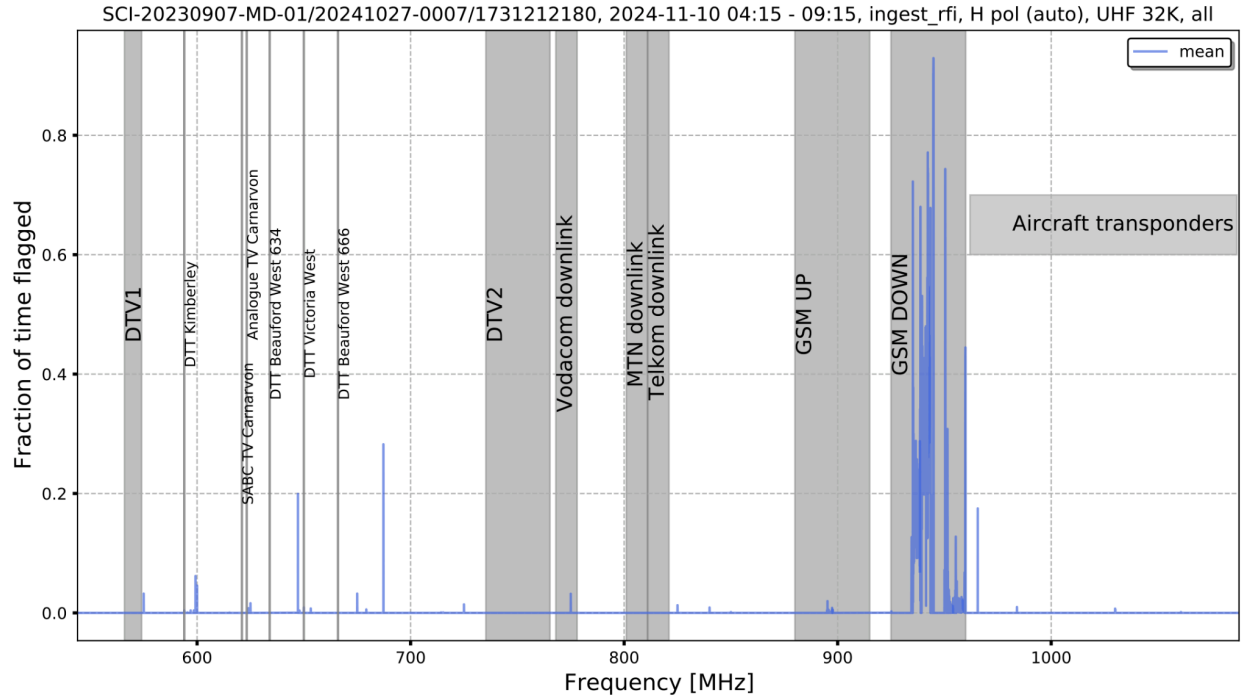


Figure 4: Mean of the autocorrelation flags of SB\_ID:20241027-0007 dataset.

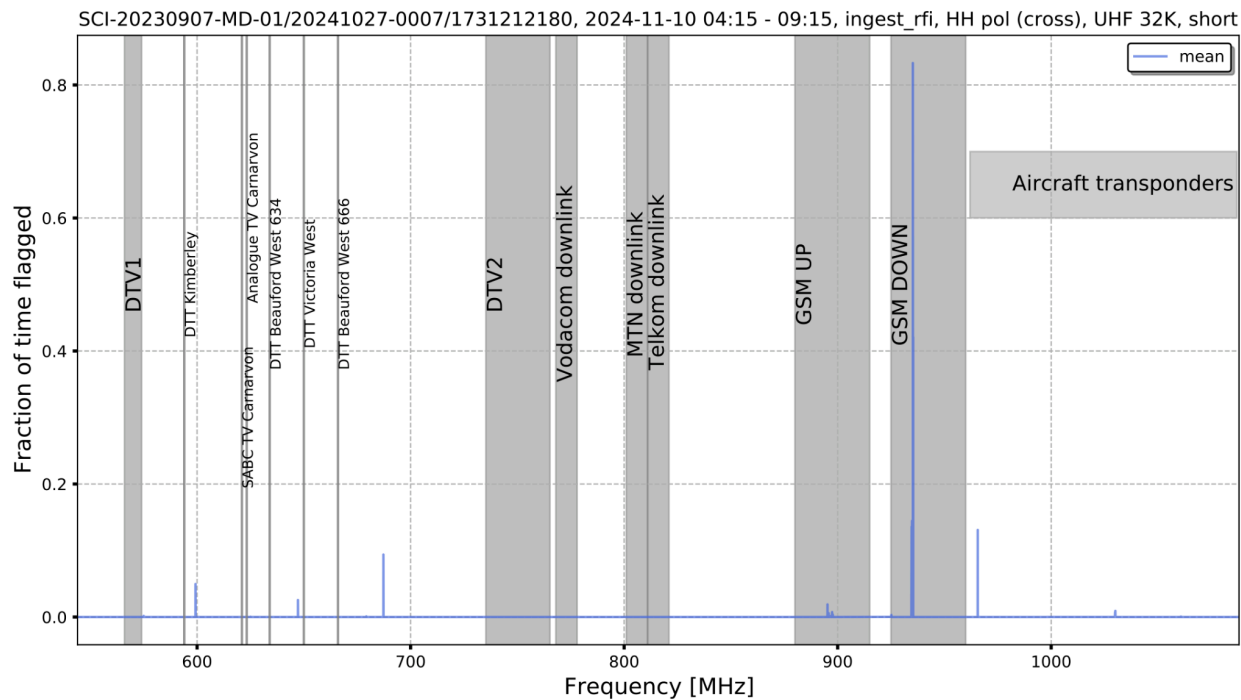


Figure 5: Mean of the cross correlation flags of the short baselines of SB\_ID:20241027-0007 dataset.

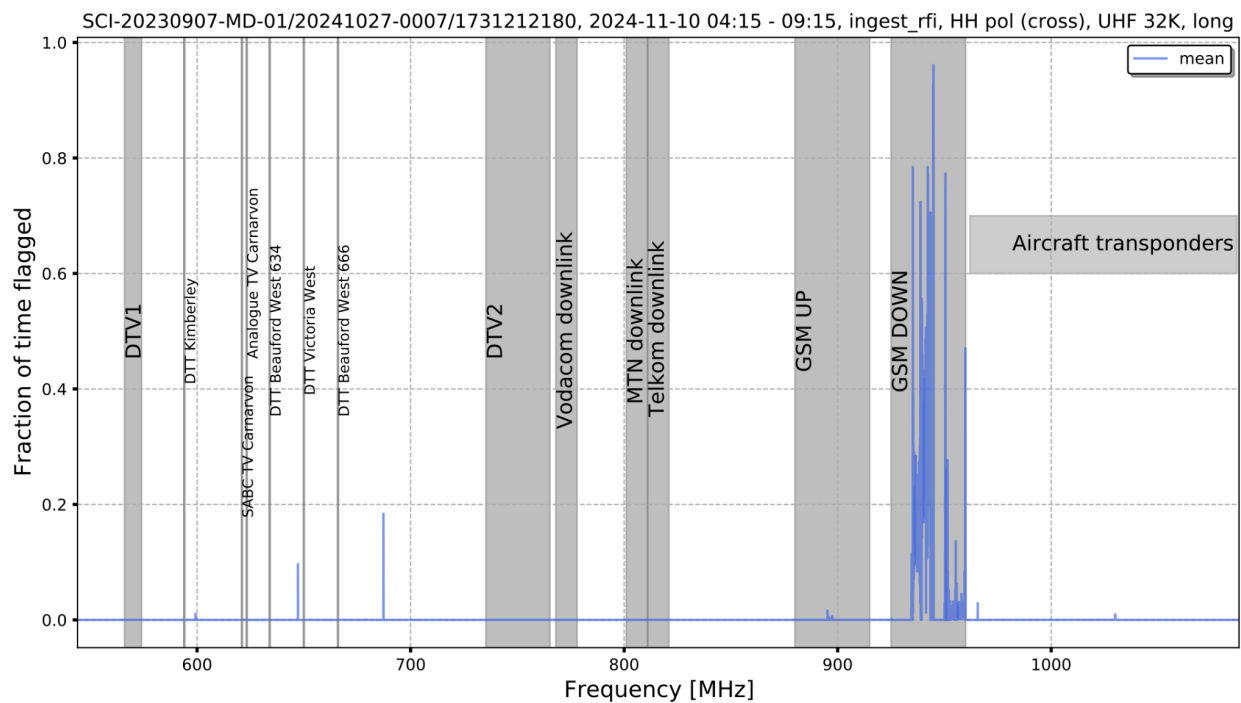


Figure 6: Mean of the cross correlation flags of long baselines of SB\_ID:20241027-0007 dataset.

**c. 2024, 17 November**

The flags presented in Figures 7, 8 and 9 are related to the dataset collected on the 17th of November. In general, the RFI emission is relatively low in all products, although the long baseline products are cleaner than the auto and short baseline products. Similar to the dataset collected on the 10th November, the contaminant at around 700 MHz is still noticeable in all products.

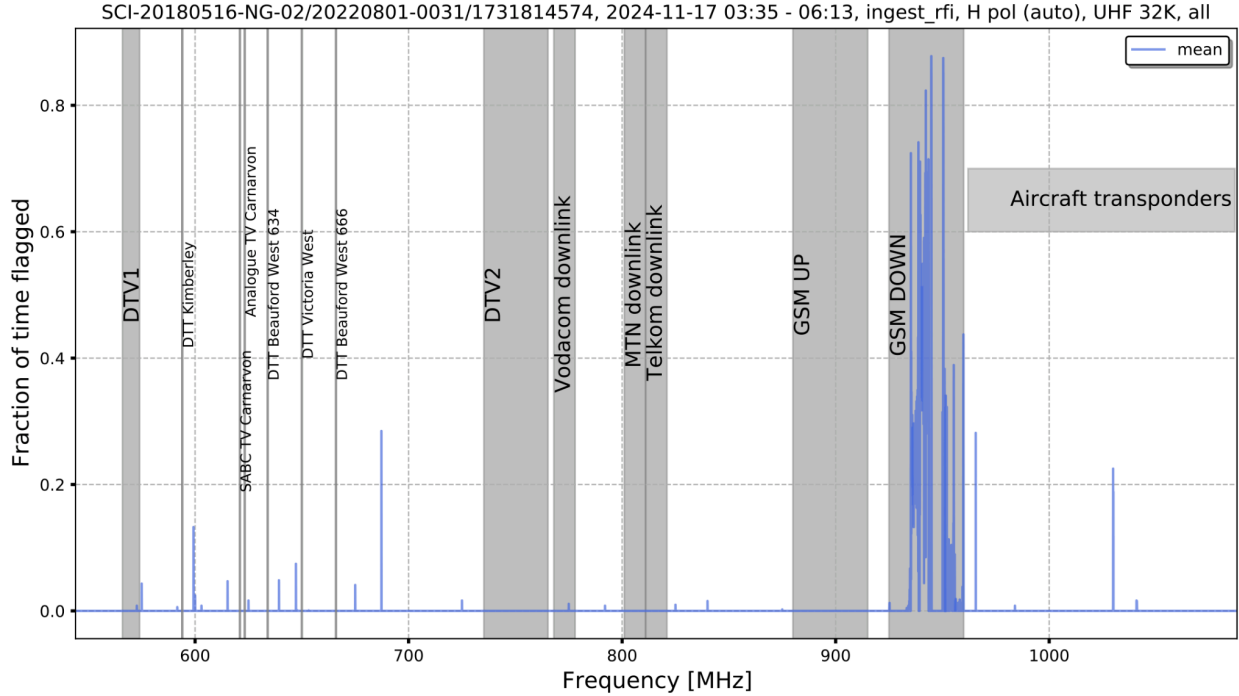


Figure 7: Mean of the autocorrelation flags of SB\_ID:20220801-0031 dataset.

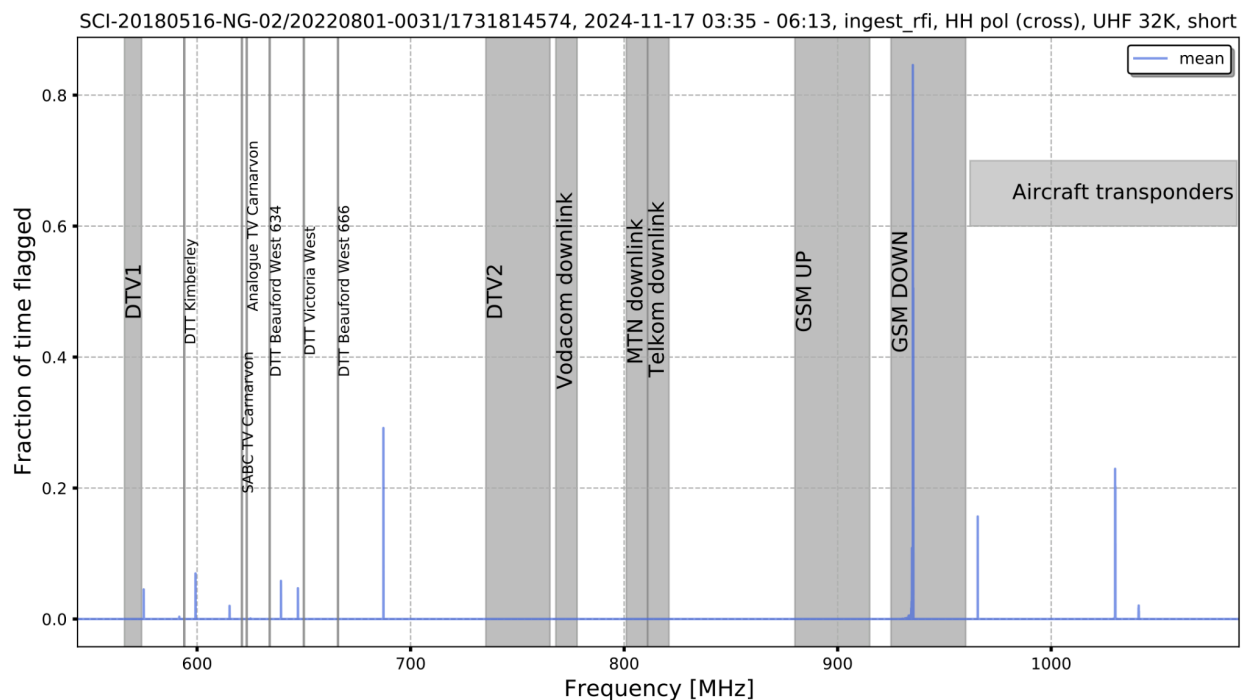


Figure 8: Mean of the cross correlation flags of the short baselines of SB\_ID:20220801-0031 dataset.

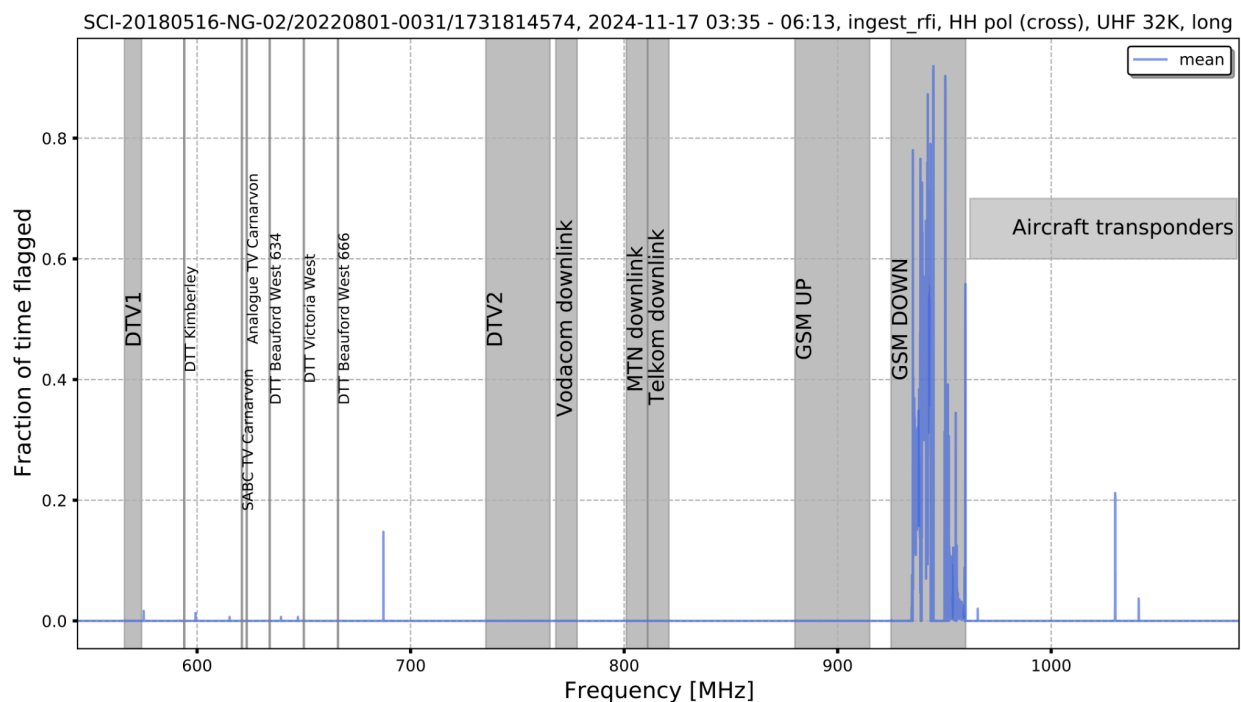


Figure 9: Mean of the cross correlation flags of the long baselines of SB\_ID:20220801-0031 dataset.

**d. 2024, 24 November**

The flags from the dataset collected on the 24th of November appear to be relatively clean as shown in Figures 10, 11 and 12. But the noise at about 700 MHz in all data products, as already mentioned in the above sections, still persists.

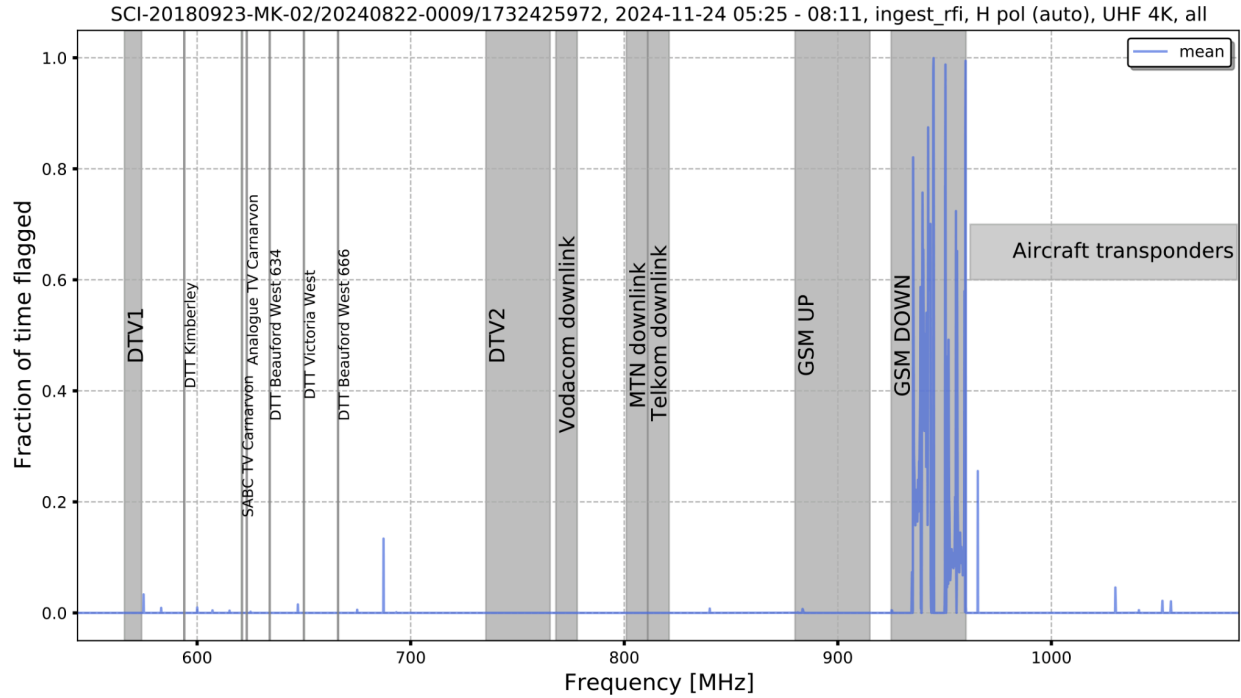


Figure 10: Mean of the autocorrelation flags of SB\_ID:20240822-0009 dataset.

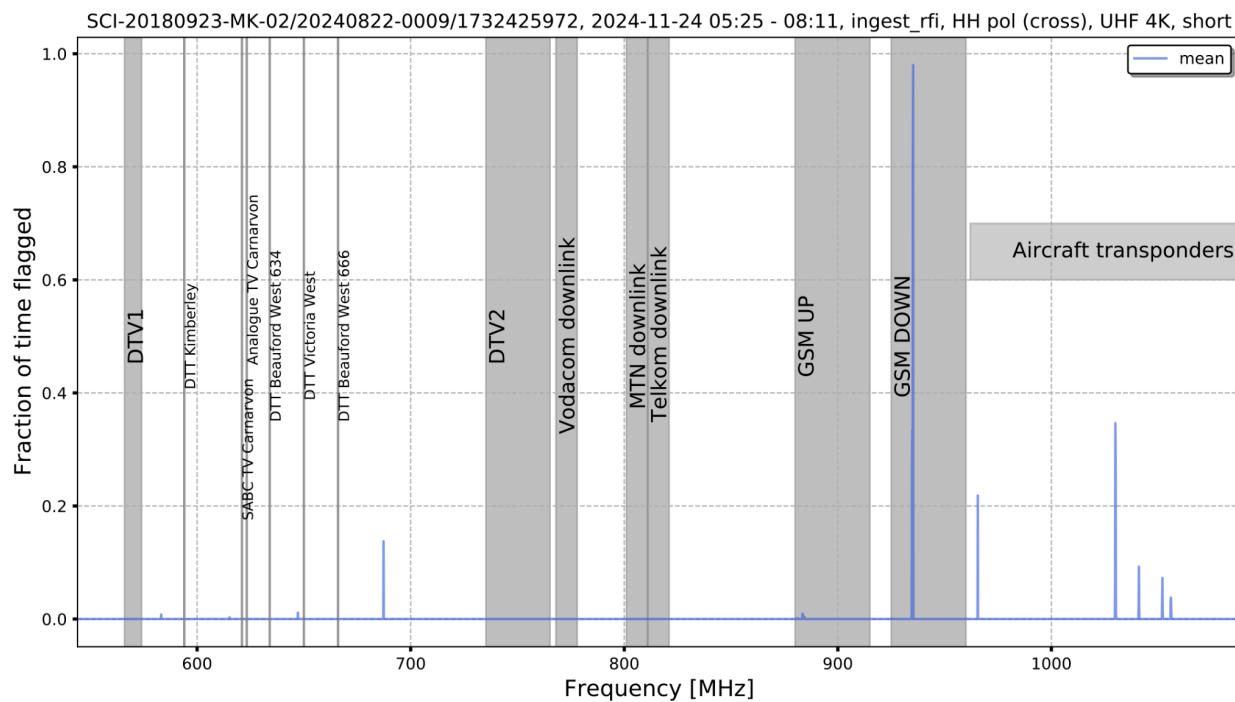


Figure 11: Mean of the cross correlation flags of the short baselines of SB\_ID:20240822-0009 dataset.

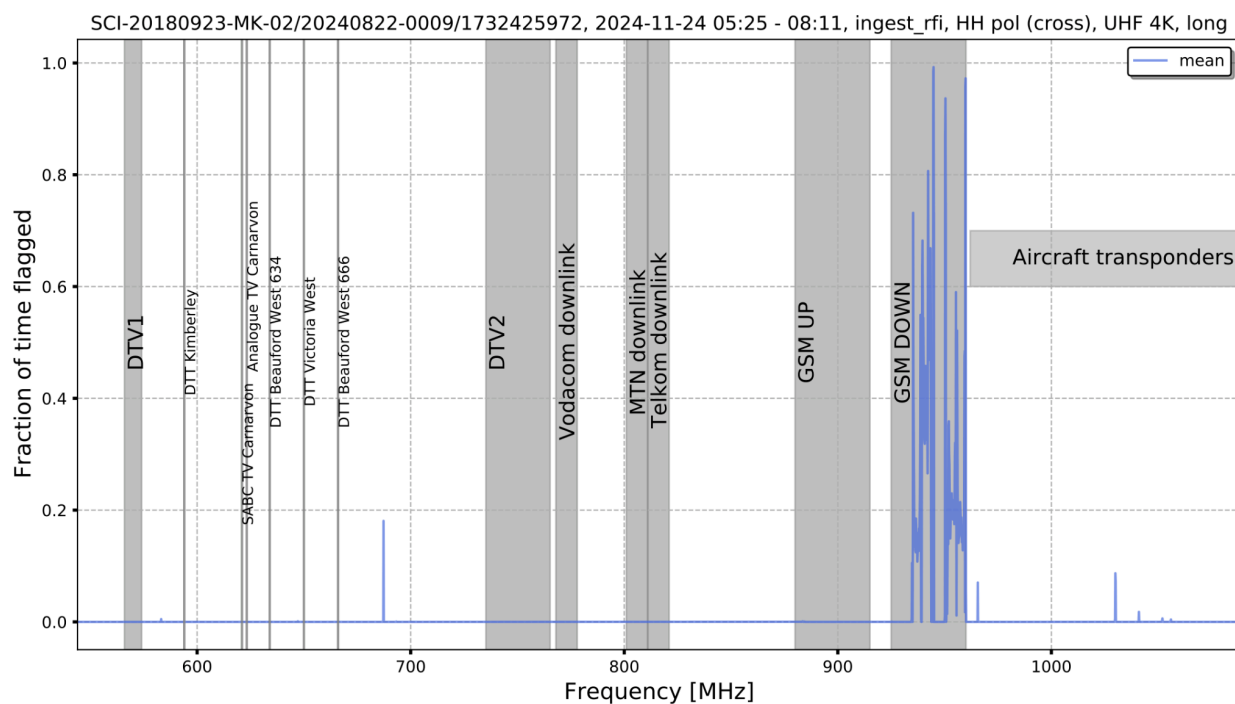


Figure 12: Mean of the cross correlation flags of the long baselines of SB\_ID:20220801-0031 dataset.

#### **4. RFI summary for November 2024 at UHF, HH cross polarization**

Now we look at the overall RFI for the month of September by concatenating all science observations for the month. Here we select particularly 8s observations, observed at 4k or 32k channel mode. The monitoring and analysis of RFI are presented using the KATHPRFI pipeline [1], highlighting the impact of construction on RFI levels.

##### **KATHPRFI algorithm**

The `kathprfi` essentially processes two numpy arrays, `Master` and `Counter`, to update the number of RFI points and observations per voxel which refers to 5-D array of Time, Frequency, Baseline, Elevation and Azimuth. It iterates over chunks of frequency indices, baseline indices, and time indices. For each combination, the algorithm updates the `Master` array by the corresponding 'Good flags' value and the `Counter` array by 1. Good flags for a given observation are based on a flag type, e.g. if flag type is calibration RFI then target tags for that observation will be obtained from a complete calibration report. The output of this is an xarray dataset that contains arrays of these two arrays and the dimensions of interest as mentioned above.

##### **Computation of marginal probability**

The RFI probability is computed by using two arrays from the output Zarr file obtained from the `kathprfi` pipeline : 'Master' and 'Counter'. These arrays represent the number of RFI points and the total number of observations, respectively. The function sums these arrays along a specified dimension to get total counts and then calculates the probability as the ratio of these sums.

RFI amount as a function of time of the day for November 2024 at UHF, HH cross pol

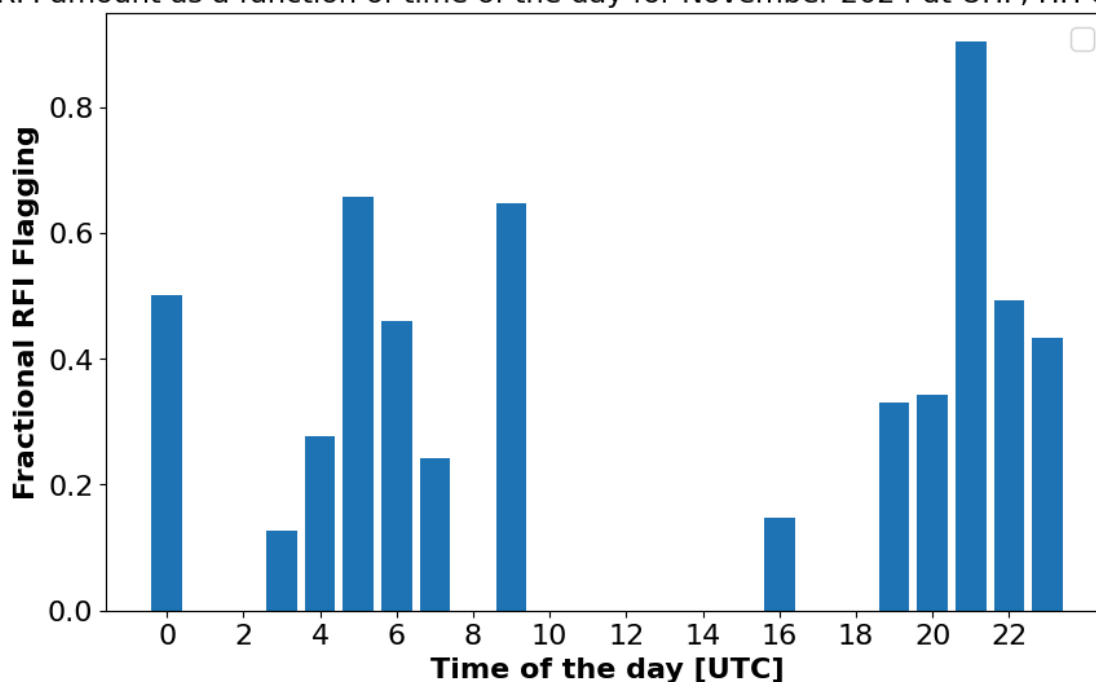


Figure 13: The plot above shows the fractional RFI flagging as a function of time of the day in UTC for November 2024 UHF horizontal cross pol observations.

Figure 13 illustrates the hourly behavior of RFI for November 2024, highlighting specific time periods affected by elevated RFI levels. The data reveals that RFI behavior is generally unstable, with an average RFI percentage of 45% throughout the day. Notably, RFI levels exceed the daily average during the hours 00:00 UTC, 05:00 UTC, 09:00 UTC, 21:00 UTC, and 22:00 UTC. Among these, a particularly high RFI fraction is observed at 21:00 UTC, where the RFI percentage peaks at 90%. This suggests a period of heightened RFI prevalence. The causes for such elevated RFI levels may include specific construction activities on-site or interference related to a source under observation during that time. To better characterize these anomalies, a detailed analysis of the frequency spectra during these hours is necessary.

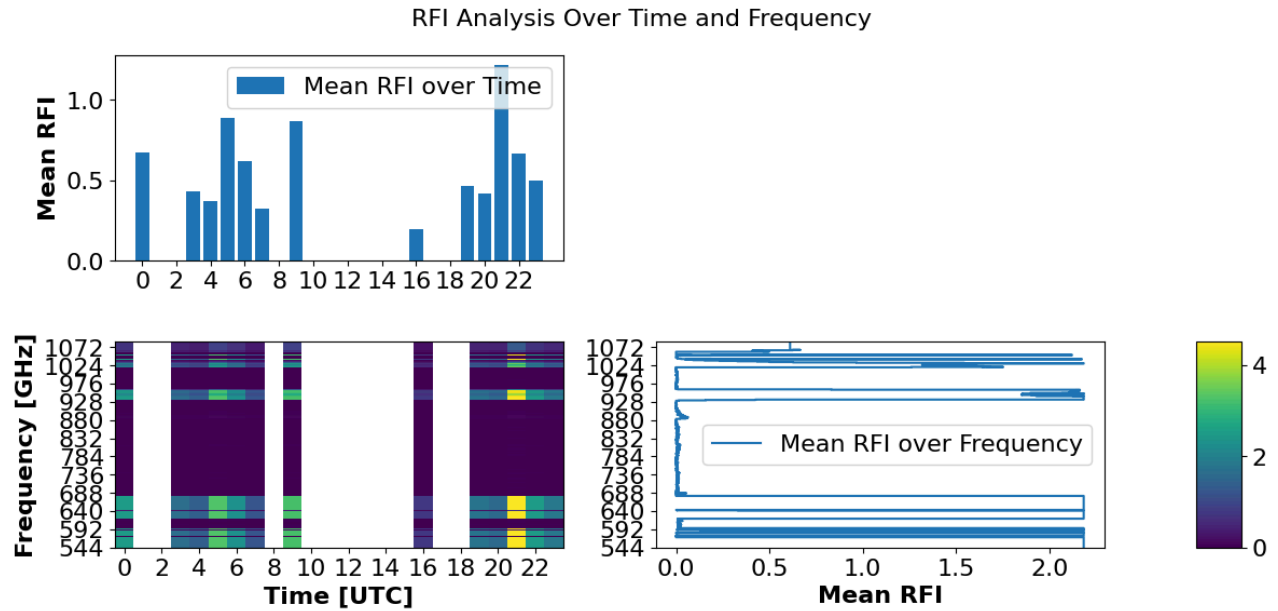


Figure 14: 2-D average RFI occupancy across time and frequency space

Figure 14 presents a two-dimensional view of RFI patterns across time and frequency. The top 1D bar plot displays mean RFI levels over time, consistent with the trends shown in Figure 13. The central 2D heatmap provides a detailed visualization of RFI activity, with white regions indicating missing data or non-monitored frequency intervals. Intense RFI zones are represented by bright yellow or green patches, as per the color bar on the far right. Significant RFI is observed in specific frequency bands, including 544–592 MHz, 620–680 MHz, 910–950 MHz, and above 1000 MHz. These correspond to peak RFI levels at 00:00 UTC (55%), 05:00 UTC (68%), 09:00 UTC (65%), and 21:00 UTC (90%). The 1D plot on the right highlights mean RFI levels across frequencies, where noticeable spikes represent strong interference at specific frequency ranges.

We recognize known RFI sources in some bands, such as GSM downlink/uplink (910–1050 MHz) and DTV1 (544–592 MHz). However, for the first time, a strong RFI signal is observed between 620–680 MHz. This range includes known sources that typically generate minimal interference, which warrants further investigation